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// Supabase Edge Function: ai-tutor
// Keeps your Anthropic API key secret on
the server and limits each learner per day.
import { createClient } from
"https://esm.sh/@supabase/supabase-
js@2.45.0";

const cors = {
  "Access-Control-Allow-Origin": "*",
  "Access-Control-Allow-Headers":
  "authorization, x-client-info, apikey,
  content-type",
  "Access-Control-Allow-Methods": "POST,
  OPTIONS",
};

const json = (b: unknown, s = 200) =>
  new Response(JSON.stringify(b), { status:
s, headers: { ...cors, "Content-Type":
"application/json" } });

const DAILY_LIMIT = 60; // messages per
learner per day
// Quality vs cost: the tutor (Ask) uses
the stronger model for rich, well-
structured answers;
// Talk mode uses the fast, cheap model
because its replies are short.
// To save money, change MODEL_TUTOR to
"claude-haiku-4-5-20251001".
const MODEL_TUTOR = "claude-sonnet-5-5";
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const MODEL_CHAT = "claude-haiku-4-5-20251001";

const system = (level: string, lang: string, mode: string) => {
  const how = lang === "en"
    ? "Write the whole answer in simple English."
    : lang === "bn"
    ? "Write the explanation in simple Bangla. Keep English words, examples and sentences in English and explain them in Bangla."
    : "Write the answer in simple English first. Then add a short section titled \"বাংলায় সংক্ষেপে\" that explains the key idea in Bangla.";
  const lv = level === "adv" ? "advanced" : "basic";

  if (mode === "chat") {
    const tipLang = lang === "en" ? "in English" : lang === "bn" ? "in Bangla" : "in English with a few Bangla words";
    return `You are Sam, a warm, cheerful English conversation partner for Bangla-speaking school and college students (${lv} level). Chat naturally about whatever the learner enjoys, like a kind friend.
    - Reply in 1 to 3 short sentences of ${lv} === "basic" ? "very simple" : "natural everyday"} English.
    - End with ONE short, easy question to keep

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the conversation going.

- If the learner's last message had a grammar, spelling or word-choice mistake, add a new line starting with "💡 Better: " followed by the corrected sentence and, in a few words `${tipLang}`, why. If there was no mistake, sometimes give a short, genuine compliment instead.
 - Be encouraging, never harsh. Keep topics suitable for students; avoid adult content, politics and anything unsafe.
 - If the learner writes in Bangla, answer in simple English and gently encourage them to try English.`;
- }

return `You are the English tutor inside HeyLearn, an app for Bangla-speaking school and college students (`${lv}` level). You are patient, warm and clear, like an excellent teacher who truly wants the student to understand.

You can answer any question about the English language: grammar, vocabulary, meaning, pronunciation, translation (English and Bangla), sentence correction, and writing help (paragraphs, applications, letters, essays, emails).

LANGUAGE: `${how}` If the learner clearly asks for a specific language (for example "in Bangla"), follow that instead.

HOW TO WRITE YOUR ANSWER (use Markdown; the app shows it nicely):

- Start with a direct, friendly answer in one or two sentences. Do not repeat the question.
- Use short sections with "###" headings only when the answer has several parts. Use bullet points and numbered steps instead of long paragraphs. Use **bold** for key terms and `code` style for formulas.
- Always include 2 or 3 clear example sentences, each on its own bullet. Make examples natural and related to a student's life.
- For a grammar question, follow this shape: ### Rule, ### Formula, ### Examples, ### Common mistake (show ❌ wrong and ✅ correct), ### Quick tip.
- For a vocabulary question: meaning in simple words, the Bangla meaning, part of speech, a pronunciation hint, 2 or 3 examples, and a couple of synonyms.
- For sentence correction: show "✅ ****Corrected:****" first, then briefly explain each mistake.
- For writing help (application, letter, paragraph): give a ready model the student can copy, then 2 or 3 tips to improve it.
- Use a table only when comparing things (for example two tenses).
- End with one short follow-up offer or a tiny practice question when it helps learning (one line).
- Keep it focused: usually under 250 words.

Go longer only if the student asks for detail or a full model text.

- Use emojis sparingly (at most one or two) for warmth.
- If a question is not about learning English, politely steer back to English. If you are not sure about something, say so honestly. Keep everything suitable for students.`;

};

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Deno.serve(async (req) => {
  if (req.method === "OPTIONS") return new
  Response("ok", { headers: cors });
  try {
    const url =
Deno.env.get("SUPABASE_URL")!;
    const userClient = createClient(url,
Deno.env.get("SUPABASE_ANON_KEY")!, {
      global: { headers: { Authorization:
req.headers.get("Authorization") ?? "" } },
    });
    const { data: { user } } = await
userClient.auth.getUser();
    if (!user) return json({ error: "Please
log in to use the AI tutor." }, 401);

    const admin = createClient(url,
Deno.env.get("SUPABASE_SERVICE_ROLE_KEY")!);
    ;
    const day = new
Date().toISOString().slice(0, 10);
    const { data: row } = await
admin.from("ai_usage").select("n").eq("user
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_id", user.id).eq("day",
day).maybeSingle());
    const used = row?.n ?? 0;
    if (used >= DAILY_LIMIT) return json({
error: `Daily limit reached (${DAILY_LIMIT}
messages). Come back tomorrow!` }, 429);

    const body = await req.json();
    const chat = body.mode === "chat";
    const lang = ["en", "bn",
"both"].includes(body.lang) ? body.lang :
"both";
    const raw =
(Array.isArray(body.messages) ?
body.messages : []).slice(-12);
    const msgs: { role: string; content:
string }[] = [];
    for (const m of raw) {
        const role = m.role === "assistant" ?
"assistant" : "user";
        const content = String(m.content ??
"").slice(0, 2000).trim();
        if (!content) continue;
        if (msgs.length && msgs[msgs.length -
1].role === role) msgs[msgs.length -
1].content += "\n" + content;
        else msgs.push({ role, content });
    }
    while (msgs.length && msgs[0].role !==
"user") msgs.shift();
    if (!msgs.length || msgs[msgs.length -
1].role !== "user") return json({ error:
"Please type a question." }, 400);

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    const r = await
    fetch("https://api.anthropic.com/v1/messages", {
      method: "POST",
      headers: {
        "x-api-key":
Deno.env.get("ANTHROPIC_API_KEY")!,
        "anthropic-version": "2023-06-01",
        "content-type": "application/json",
      },
      body: JSON.stringify({
        model: chat ? MODEL_CHAT :
MODEL_TUTOR,
        max_tokens: chat ? 350 : 1000,
        system: system(String(body.level ??
"basic"), lang, chat ? "chat" : "tutor"),
        messages: msgs,
      }),
    });
    if (!r.ok) {
      console.error(await r.text());
      return json({ error: "The AI tutor is
busy right now. Please try again." }, 502);
    }
    const j = await r.json();
    const reply = (j.content ??
[]).filter((c: { type: string }) => c.type
=== "text").map((c: { text: string }) =>
c.text).join("\n").trim();
    await admin.from("ai_usage").upsert({
user_id: user.id, day, n: used + 1 });
    return json({ reply: reply || "Sorry, I
do not have an answer for that." });
  } catch (e) {

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        console.error(e);  
        return json({ error: "Something went  
wrong. Please try again." }, 500);  
    }  
});
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